Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 1 — Algebraic methods
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第1章 · 代数方法
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C01-R01 · P2 · Chapter 1

$f(x) = 2x^3 - 3x^2 - 11x + 6.$

(a) Show that $(x - 3)$ is a factor of $f(x)$. (2)

(b) Factorise $f(x)$ completely and hence solve $f(x) = 0$. (4)

(Total for Question 1 is 6 marks)

$f(x) = x^3 + ax + b$. When divided by $(x - 1)$ the remainder is 4, and when divided by $(x + 2)$ the remainder is -11.

(a) Find a, b. (3)

(b) Find the remainder when $f(x)$ is divided by $(2x - 1)$. (2)

(Total for Question 2 is 5 marks)

3. P2-C01-R03 · P2 · Chapter 1

$f(x) = x^4 - 5x^2 + 4$.

(a) Factorise $f(x)$ completely. (3)

(b) Find all real solutions of $f(x) = 4$. (2)

(Total for Question 3 is 5 marks)

4.

Let $f(x) = x^3 + px^2 + qx - 12$. Both $(x - 2)$ and $(x + 3)$ are factors.

(a) Find p, q. (3)

(b) Find the remaining linear factor. (2)

(Total for Question 4 is 5 marks)

5.

For an integer n, consider $N = n^2 + n + 1$.

(a) Prove that N is odd for every integer n. (3)

(b) Disprove the claim that N is prime for every positive integer n. (2)

(Total for Question 5 is 5 marks)

$f(x) = 3x^3 + 2x^2 - 5x + 7.$

(a) Find the quotient and remainder when $f(x)$ is divided by $(x + 1)$. (3)

(b) Find the value of $f(-1)$ without substituting into the original cubic, and state the theorem used. (2)

(Total for Question 6 is 5 marks)

7.

A polynomial f leaves remainder 5 on division by $(x-2)$. Define $g(x) = f(x) + 3x - 1$.

(a) Find the remainder when $g(x)$ is divided by $(x-2)$. (2)

(b) Find k such that $(x-2)$ is a factor of $f(x) + kx$. (2)

(Total for Question 7 is 4 marks)

8. P2-C01-R08 · P2 · Chapter 1

$f(x) = x^3 - 4x^2 + x + 6.$

(a) Show that $f(x) = (x + 1)(x - 2)(x - 3)$. (3)

(b) Find the roots of $f(2t - 1) = 0$. (2)

(Total for Question 8 is 5 marks)

9. P2-C01-R09 · P2 · Chapter 1

An integer n satisfies $1 \leq n \leq 6$.

(a) By exhaustion, prove that $n^2 + 2$ is divisible by 3 if and only if n is not divisible by 3, for these six integers. (4)

(b) Give a reason why checking only $n = 1, 2, 3$ would not by itself be a proof by exhaustion over the stated domain. (2)

(Total for Question 9 is 6 marks)

The polynomial $f(x) = 2x^3 + ax^2 + bx - 6$ is divisible by $(x - 1)(x + 2)$.

(a) Find a, b. (3)

(b) Write $f(x)$ as a product of three linear factors. (2)

(Total for Question 10 is 5 marks)

11. P2-C01-R11 · P2 · Chapter 1

For real x, $F(x) = (x-2)^3 - (x-2)$.

(a) Factorise $F(x)$ completely. (3)

(b) Find all real values of x for which $F(x) = 0$, and show that their sum is 6. (2)

(Total for Question 11 is 5 marks)

$f(x) = x^3 + 2x^2 + cx + d$. When divided by $(x - 1)$ and $(x + 1)$, the remainders are equal.

(a) Show that $c = -1$. (3)

(b) Given that $(x - 2)$ is a factor, find d and the common remainder. (4)

(Total for Question 12 is 7 marks)

13.

A student claims that if a polynomial has equal values at $x = 1$ and $x = -1$, it must contain only even powers of x.

(a) Give a polynomial counterexample of degree 3. (3)

(b) Find the remainder when your polynomial is divided by $x^2 - 1$. (2)

(Total for Question 13 is 5 marks)

$f(x) = x^3 - 2x^2 - 5x + 6.$

(a) Divide $f(x)$ by $x - 1$ and hence factorise it completely. (4)

(b) Find the exact roots of $f(\sqrt{t}) = 0$ for $t \geq 0$. (3)

(Total for Question 14 is 7 marks)

$f(x) = ax^3 + bx^2 + cx + d$ is identically equal to $(x + 1)(2x^2 - 3x + 4) + 5$.

(a) Find a, b, c, d. (3)

(b) Find $f(-1)$ and the remainder on division by $x - 2$. (3)

(Total for Question 15 is 6 marks)

16. P2-C01-R16 · P2 · Chapter 1

For integer n, let $E = (2n+1)^2 - 1$.

(a) Prove that E is divisible by 8. (3)

(b) Is E necessarily divisible by 16? Justify your answer with a specific integer value. (2)

(Total for Question 16 is 5 marks)

$P(x) = x^4 + ax^2 + b$. Its remainders on division by $x - 1$ and $x - 2$ are 0 and 9, respectively.

(a) Find a, b. (3)

(b) Find the quotient when P is divided by $x^2 - 1$. (2)

(Total for Question 17 is 5 marks)

The polynomial $f(x)$ has $(x + 1)$ as a factor. Let $g(x) = f(2x - 3)$.

(a) Find a linear factor of $g(x)$. (3)

(b) Given also that $f(3) = 8$, find the remainder when g is divided by $x - 3$. (2)

(Total for Question 18 is 5 marks)

A cubic polynomial with leading coefficient 2 has roots -1, 2 and 3.

(a) Find the polynomial in expanded form. (3)

(b) Find the remainder on division by $2x + 1$. (3)

(Total for Question 19 is 6 marks)

20.

Consider the statement: “For every real x, $x^2 \geq x$.”

(a) Find the complete set of values of x for which the statement is true. (2)

(b) Give a counterexample to the statement and explain why checking only integer values would miss every counterexample. (3)

(Total for Question 20 is 5 marks)

$f(x) = 4x^3 - 4x^2 - 11x + 6.$

(a) Show that $(2x - 1)$ is a factor and find the quotient. (3)

(b) Hence solve $f(x) = 0$ exactly. (2)

(Total for Question 21 is 5 marks)

A polynomial $P(x)$ satisfies the identity $P(x) = (x - 2)Q(x) + 3x + 1$.

(a) Find the remainder when P is divided by $x - 2$. (2)

(b) Write the identity in division form with a constant remainder. (3)

(Total for Question 22 is 5 marks)

For real x, $H(x) = x^3 + 3x^2 - 4$.

(a) Show that $H(x) = (x - 1)(x + 2)^2$. (3)

(b) Find the number of distinct real solutions of $H(x) = 0$ and state them. (2)

(Total for Question 23 is 5 marks)

$f(x) = x^3 + ax^2 + bx + 4$. The remainder on division by $x - 1$ is twice the remainder on division by $x + 1$. Also $f(2) = 12$.

(a) Find a, b. (4)

(b) Find the remainder on division by x. (2)

(Total for Question 24 is 6 marks)

For integers $m, n,$ consider the claim: “If mn is even, then both m and n are even.”

(a) Disprove the claim. (3)

(b) Prove that if m and n are both odd, then mn is odd. (3)

(Total for Question 25 is 6 marks)

$f(x) = x^4 - 2x^3 - 7x^2 + 8x + 12.$

(a) Divide $f(x)$ by $x^2 - 4$. (3)

(b) Hence factorise $f(x)$ completely. (3)

(Total for Question 26 is 6 marks)

A cubic polynomial f satisfies $f(0) = 0$, $f(1) = 0$, $f(2) = 0$ and $f(3) = 12$.

(a) Find $f(x)$. (3)

(b) Find the remainder when $f(x)$ is divided by $x + 1$. (2)

(Total for Question 27 is 5 marks)

Let n be an integer. Consider $n^3 - n$.

(a) Prove that $n^3 - n$ is divisible by 6. (4)

(b) Give a counterexample to the stronger claim that $n^3 - n$ is always divisible by 12. (2)

(Total for Question 28 is 6 marks)

$f(x) = 2x^3 - 7x^2 + 4x + 4$. It is known that $(x - 2)$ is a factor.

(a) Find the quotient on division by $x - 2$ and show that $x = 2$ is a repeated root. (3)

(b) Find all distinct real roots and the remainder on division by $x + 2$. (4)

(Total for Question 29 is 7 marks)

The remainder on dividing $f(x)$ by $x-1$ is 3, and by $x+1$ is 7. Let the remainder on division by x^2-1 be $ax+b$.

(a) Find a, b. (4)

(b) Find the remainder when $f(x)+2x-5$ is divided by x^2-1. (2)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C01-R01 · 6 marks

(a) · 2 marks

$f(3) = 54 - 27 - 33 + 6 = 0.$

按因式定理检验指定零点；必须展示代入后为0。

M1	正确代入3
A1	算得0并引用因式定理

(b) · 4 marks

$f(x) = (x-3)(2x-1)(x+2)$; roots $3, 1/2, -2$.

除x−3得$2x^2+3x-2$，再分解(2x−1)(x+2)。

M1	正确商式
M1	分解二次式
A1	完全因子
A1	三根

Question 2

P2-C01-R02 · 5 marks

(a) · 3 marks

$a=2, b=1$.

f(1)=4给a+b=3；f(−2)=−11给−2a+b=−3，联立。

M1	两个余式条件
M1	联立消元
A1	a与b

(b) · 2 marks

$17/8$.

代x=1/2得1/8+1+1。

M1	用线性除式零点
A1	精确余式

Question 3

P2-C01-R03 · 5 marks

(a) · 3 marks

$(x-2)(x+2)(x-1)(x+1)$.

先令u=x^2分解$(u-1)(u-4)$，再两次平方差。

M1	关于x平方分解
M1	平方差
A1	四个一次因子

(b) · 2 marks

$x=0,\pm\sqrt{5}$.

原式减4后$x^2(x^2-5)=0$；零根只计一次。

M1	正确移项并因式分解
A1	三实根

Question 4

P2-C01-R04 · 5 marks

(a) · 3 marks

$p = 3, q = -4.$

f(2)=0给2p+q=2；f(−3)=0给3p−q=13；相加5p=15，故p=3、q=−4。

M1	两因式条件
M1	解联立
A1	两参数

(b) · 2 marks

$(x + 2).$

常数−12及前两因子乘积(x−2)(x+3)常数−6，第三因子常数2，首项系数1。

M1	用常数或多项式除法
A1	剩余因子

Question 5

P2-C01-R05 · 5 marks

(a) · 3 marks

$n(n+1)$ is even, so $N = n(n+1) + 1$ is odd.

连续两整数必一偶，乘积为偶；加1为奇。

M1	连续整数中有偶数
M1	乘积为偶
A1	奇数结论

(b) · 2 marks

$n = 4$ gives $N = 21 = 3 \times 7$.

一个满足正整数条件且得合数的反例足以否定全称命题。

M1	合法反例值
A1	展示合数分解

Question 6

P2-C01-R06 · 5 marks

(a) · 3 marks

Quotient $3x^2 - x - 4$; remainder 11.

长除法商如上；展开$(x+1)(3x^2-x-4)+11$复核原式。

M1	正确长除法步骤
A1	商
A1	余式

(b) · 2 marks

$f(-1) = 11$ by the remainder theorem.

在除法恒等式中$x=-1$使除式项为0，只剩余式。

M1	从除法式读取
A1	值和定理

Question 7

P2-C01-R07 · 4 marks

(a) · 2 marks

10.

$f(2)=5$，$g(2)=5+6-1=10$。

M1	使用$f(2)$条件
A1	g余式

(b) · 2 marks

$k=-5/2$.

新多项式在2处为$5+2k$，要因式故设0。

M1	因式条件
A1	k

Question 8

P2-C01-R08 · 5 marks

(a) · 3 marks

$(x-2)(x-3)=x^2-5x+6$; multiplying by $x+1$ gives $f(x)$.

直接展开证明恒等式；不能仅列三个数为零而不说明首项和次数。

M1	展开两个因子
M1	乘第三因子
A1	与原式一致

(b) · 2 marks

$t=0,3/2,2$.

输入2t−1必须等于原根−1、2、3，分别解一次方程。

M1	正确三个输入条件
A1	全部t值

Question 9

P2-C01-R09 · 6 marks

(a) · 4 marks

$n^2+2=3,6,11,18,27,38$ for $n=1,2,3,4,5,6$; divisible at $1,2,4,5$ only.

穷举所有六个允许值，列结果并与n被3整除情况逐一比较。

M1	列齐六种情况
A1	六个值
M1	检查整除
A1	双向结论

(b) · 2 marks

The permitted cases $n=4,5,6$ would remain unchecked.

有限域穷举必须覆盖每一种允许情况；本题未另证明周期推广。

M1	明确遗漏哪些情况
A1	穷举覆盖理由

Question 10

P2-C01-R10 · 5 marks

(a) · 3 marks

$a=5, b=-1.$

f(1)=0给a+b=4；f(−2)=0给4a−2b=22；解a=5,b=−1。

M1	两零点条件
M1	联立
A1	参数

(b) · 2 marks

$(x-1)(x+2)(2x+3).$

先两因子乘x^2+x-2，商为2x+3；展开复核。

M1	正确商因子
A1	完整三因子

Question 11

P2-C01-R11 · 5 marks

(a) · 3 marks

$(x-2)(x-3)(x-1)$.

提x−2后余因子$(x-2)^2-1$，再用平方差。

M1	提公共因子
M1	平方差
A1	完整分解

(b) · 2 marks

$x=1,2,3$; sum 6.

各一次因子分别为0，求和。

M1	三根
A1	全部值及和

Question 12

P2-C01-R12 · 7 marks

(a) · 3 marks

$f(1)-f(-1)=2+2c=0\Rightarrow c=-1.$

两个余式分别3+c+d与1−c+d，相等后d抵消。

M1	两余式条件
M1	正确相减
A1	c

(b) · 4 marks

$d=-14$; common remainder -12.

f(2)=8+8−2+d=0给d=−14；f(1)=3−1−14=−12。

M1	因式条件
A1	d
M1	回代求余式
A1	余式

Question 13

P2-C01-R13 · 5 marks

(a) · 3 marks

$f(x) = x^3 - x$ has $f(1) = f(-1) = 0$ but contains odd powers.

提供一个三次多项式并逐一验证两个值相同，同时指出奇次项仍存在。

M1	合法三次多项式
M1	验证相等值
A1	指出违反结论

(b) · 2 marks

0 for $f(x) = x^3 - x = x(x^2 - 1)$.

把(a)中的明确例子作分解，除式是其因式。

M1	使用自己的同一个例子
A1	余式零

Question 14

P2-C01-R14 · 7 marks

(a) · 4 marks

Quotient x^2-x-6; $f=(x-1)(x-3)(x+2)$.

长除法得二次商且余式0，再分解二次商。

M1	正确长除法
A1	商和零余式
M1	分解商
A1	三因子

(b) · 3 marks

$t=1,9$.

$\sqrt{\text{t}}$只能非负，原根-2须排除；输入为1或3后平方得到t。

M1	使用非负输入限制
M1	根映射回t
A1	完整集合

Question 15

P2-C01-R15 · 6 marks

(a) · 3 marks

$a = 2, b = -1, c = 1, d = 9.$

展开乘积得$2x^3-x^2+x+4$，再加5；比较系数。

M1	正确展开
M1	常数相加
A1	四系数

(b) · 3 marks

$f(-1) = 5$; remainder 23.

由恒等式x=−1使乘积为0；x=2时3(8−6+4)+5=23。

B1	f负1
M1	代2求余式
A1	余式

Question 16

P2-C01-R16 · 5 marks

(a) · 3 marks

$E = 4n(n+1)$ and $n(n+1)$ is even.

展开后提4，连续整数乘积含2，因此整体含8。

M1	展开因式分解
M1	连续整数偶性
A1	整除8

(b) · 2 marks

No; $n = 1$ gives $E = 8$.

选允许整数1，得到8不被16整除，构成反例。

M1	合法整数例
A1	明确不整除16

Question 17

P2-C01-R17 · 5 marks

(a) · 3 marks

$a=-2, b=1.$

1+a+b=0，16+4a+b=9；相减15+3a=9给a=−2,b=1。

M1	两余式方程
M1	联立
A1	两参数

(b) · 2 marks

$x^2-1.$

$P=x^4-2x^2+1=(x^2-1)^2$。

M1	识别完全平方
A1	商

Question 18

P2-C01-R18 · 5 marks

(a) · 3 marks

$(x-1)$.

使输入$2x-3=-1$，得$x=1$，故$g(1)=0$。

M1	映射已知根
M1	因式定理
A1	因式

(b) · 2 marks

8.

$g(3)=f(6-3)=f(3)=8$。

M1	正确复合输入
A1	余式

Question 19

P2-C01-R19 · 6 marks

(a) · 3 marks

$2x^3-8x^2+2x+12$.

先写2(x+1)(x−2)(x−3)，展开。

M1	正确三个因子及系数
M1	展开
A1	正确四项

(b) · 3 marks

$35/4$.

除式零点−1/2，因子式代入得2(1/2)(−5/2)(−7/2)=35/4。

M1	使用正确除式根
M1	代入因式式
A1	余式

Question 20

P2-C01-R20 · 5 marks

(a) · 2 marks

$x \leq 0$ or $x \geq 1$.

$x^2-x=x(x-1)\geqslant 0$，取两根外侧。

M1	建立不等式并分解
A1	完整解集

(b) · 3 marks

$x = 1/2$ gives $1/4 < 1/2$; all counterexamples lie in $(0, 1)$, which contains no integers.

引用(a)的补集解释为何整数检验不足；不仅只说检查不充分。

M1	合法反例及计算
M1	确定反例区间
A1	整数检验局限

Question 21

P2-C01-R21 · 5 marks

(a) · 3 marks

$f(1/2)=0$; quotient $2x^2-x-6$.

代入1/2检验；长除2x−1得到二次式。

M1	正确零点检查
M1	长除法
A1	商及零余式

(b) · 2 marks

$x=1/2, 2, -3/2$.

二次商(2x+3)(x−2)，和已知因子共同给三根。

M1	正确分解商
A1	全部三根

Question 22

P2-C01-R22 · 5 marks

(a) · 2 marks

7.

当前3x+1次数不小于除式，不能直接当余式；令x=2求P(2)=7。

M1	正确代除式根
A1	余式7

(b) · 3 marks

$P(x) = (x-2)(Q(x)+3)+7.$

把3x+1重写成3(x−2)+7，再合并商。

M1	重写线性剩余项
M1	合并商
A1	合法除法形式

Question 23

P2-C01-R23 · 5 marks

(a) · 3 marks

$(x+2)^2 = x^2 + 4x + 4$; multiplying by $x - 1$ gives $x^3 + 3x^2 - 4$.

逐项展开，注意x项抵消。

M1	展开平方
M1	乘法并合并
A1	目标式

(b) · 2 marks

Two: $x = -2, 1$.

平方因子给一个不同的根-2，不能把重根算作两个不同解。

M1	从因子找根
A1	两个不同解及数量

Question 24

P2-C01-R24 · 6 marks

(a) · 4 marks

$a=-\frac{1}{7}, b=\frac{2}{7}$.

5+a+b=2(3+a−b)给−a+3b=1；f(2)=12给2a+b=0；解a=−1/7,b=2/7。

M1	正确余式倍数关系
M1	第二条件
M1	解联立
A1	参数

(b) · 2 marks

4.

除x的余式为f(0)，就是常数项，不依赖参数。

M1	使用零点0
A1	余式

Question 25

P2-C01-R25 · 6 marks

(a) · 3 marks

$m = 2, n = 3$ gives even product 6, but n is odd.

反例必须让前提成立同时结论失败。

M1	正确反例
M1	验证前提
A1	验证结论失败

(b) · 3 marks

$m = 2r + 1, n = 2s + 1 \Rightarrow mn = 2(2rs + r + s) + 1$.

把两奇数写成2整数加1，展开并提出2，括号仍为整数。

M1	正确奇数代数表示
M1	展开整理
A1	奇数结论

Question 26

P2-C01-R26 · 6 marks

(a) · 3 marks

Quotient x^2-2x-3, remainder 0.

长除法或展开$(x^2-4)(x^2-2x-3)$复核。

M1	正确首项除法
M1	后续相减
A1	商和余式

(b) · 3 marks

$(x-2)(x+2)(x-3)(x+1)$.

除式平方差，商分解$(x-3)(x+1)$。

M1	分解除式
M1	分解商
A1	完全因式

Question 27

P2-C01-R27 · 5 marks

(a) · 3 marks

$f(x) = 2x(x-1)(x-2)$.

三个已知根给f=kx(x−1)(x−2)，代3得6k=12。

M1	因式定理构造
M1	用非零函数值定k
A1	多项式

(b) · 2 marks

-12.

f(−1)=2(−1)(−2)(−3)=−12。

M1	代正确除式根
A1	余式

Question 28

P2-C01-R28 · 6 marks

(a) · 4 marks

$n^3-n=(n-1)n(n+1)$; among three consecutive integers one is divisible by 3 and at least one is even.

连续三整数积含2也含3，2和3互素，所以含6；对负整数和零同样成立。

M1	正确三因子
M1	整除2理由
M1	整除3理由
A1	整除6结论

(b) · 2 marks

$n=2$ gives 6, not divisible by 12.

列正整数反例并计算，不能只说6不一定是12倍数。

M1	反例n及数值
A1	明确失败

Question 29

P2-C01-R29 · 7 marks

(a) · 3 marks

Quotient $2x^2-3x-2=(2x+1)(x-2)$, so $(x-2)^2$ divides f.

长除得二次商，再检查商也在2为0或直接分解。

M1	正确商
M1	二次商分解
A1	重因式结论

(b) · 4 marks

Roots $2, -1/2$; remainder -48.

f=(x−2)²(2x+1)给两不同根；代−2得16×(−3)=−48。

M1	正确不同根
A1	两根
M1	余式代值
A1	余式

Question 30

P2-C01-R30 · 6 marks

(a) · 4 marks

$a=-2, b=5$.

除法式在±1给a+b=3、−a+b=7，联立。

M1	设合法一次余式
M1	两个条件
M1	解联立
A1	a与b

(b) · 2 marks

0.

加上的线性式正好抵消已求余式−2x+5，所以可整除。

M1	正确相加余式
A1	零余式